Step 1: Import Required Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
```

Step 2: Load Dataset

```
df = pd.read_csv("/content/Global_Superstore(CSV).csv")
df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Code	City	...	Product ID	Category
0	40098	CA-2014-AB10015140-41954	11/11/2014	11/13/2014	First Class	AB-100151402	Aaron Bergman	Consumer	73120.0	Oklahoma City	...	TEC-PH-5816	Technology
1	26341	IN-2014-JR162107-41675	2/5/2014	2/7/2014	Second Class	JR-162107	Justin Ritter	Corporate	NaN	Wollongong	...	FUR-CH-5379	Furniture
2	25330	IN-2014-CR127307-41929	10/17/2014	10/18/2014	First Class	CR-127307	Craig Reiter	Consumer	NaN	Brisbane	...	TEC-PH-5356	Technology
3	13524	ES-2014-KM1637548-41667	1/28/2014	1/30/2014	First Class	KM-1637548	Katherine Murray	Home Office	NaN	Berlin	...	TEC-PH-5267	Technology
4	47221	SG-2014-RH9495111-41948	11/5/2014	11/6/2014	Same Day	RH-9495111	Rick Hansen	Consumer	NaN	Dakar	...	TEC-CO-6011	Technology

5 rows × 24 columns

Convert the date column:

```
df["Order Date"] = pd.to_datetime(df["Order Date"])
```

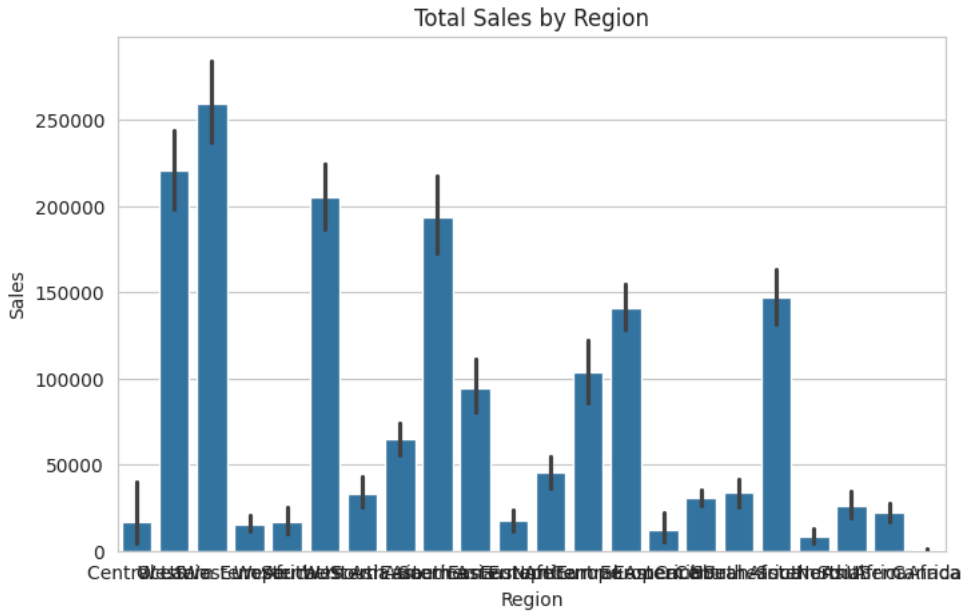
Step 3: Bar Chart (Sales by Region)

```
plt.figure(figsize=(8,5))

sns.barplot(data=df, x="Region", y="Sales", estimator=sum)

plt.title("Total Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

plt.show()
```



Step 4: Line Chart (Monthly Sales Trend)

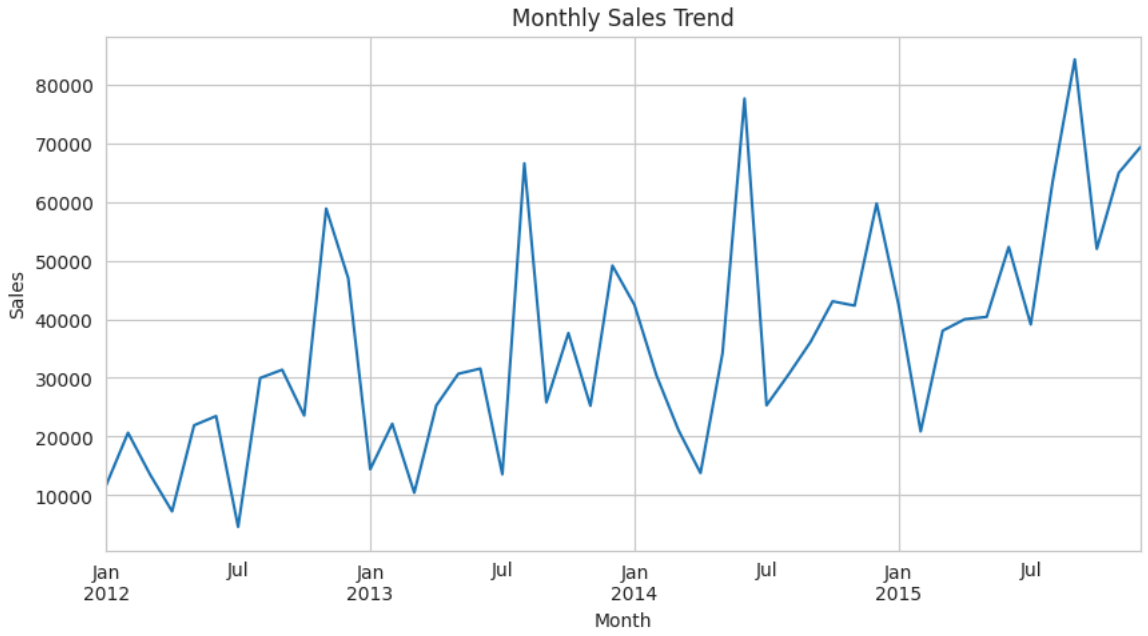
```
monthly_sales = df.groupby(df["Order Date"].dt.to_period("M"))["Sales"].sum()

plt.figure(figsize=(10,5))

monthly_sales.plot()

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.show()
```



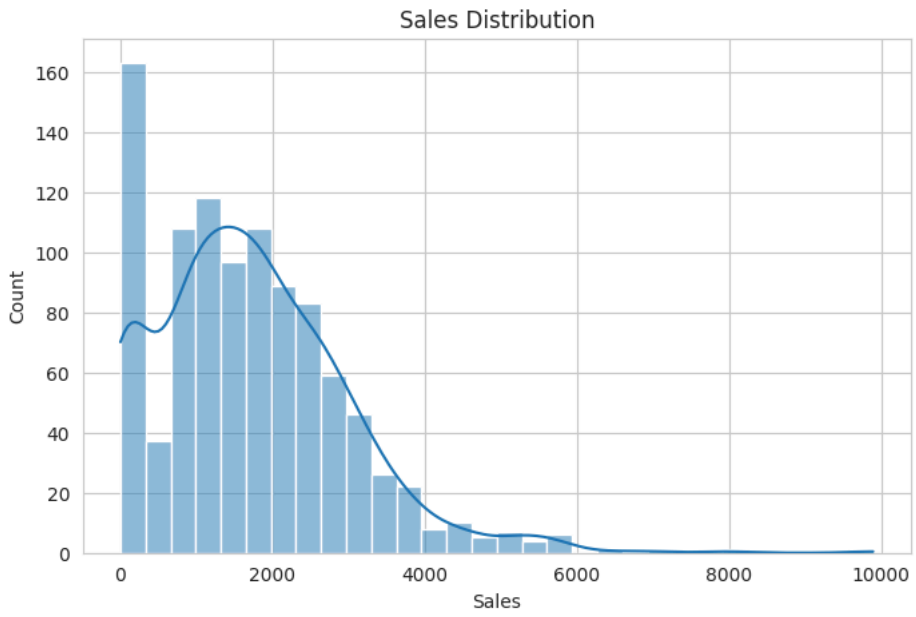
Step 5: Histogram (Sales Distribution)

```
plt.figure(figsize=(8,5))

sns.histplot(df["Sales"], bins=30, kde=True)

plt.title("Sales Distribution")
```

```
plt.show()
```



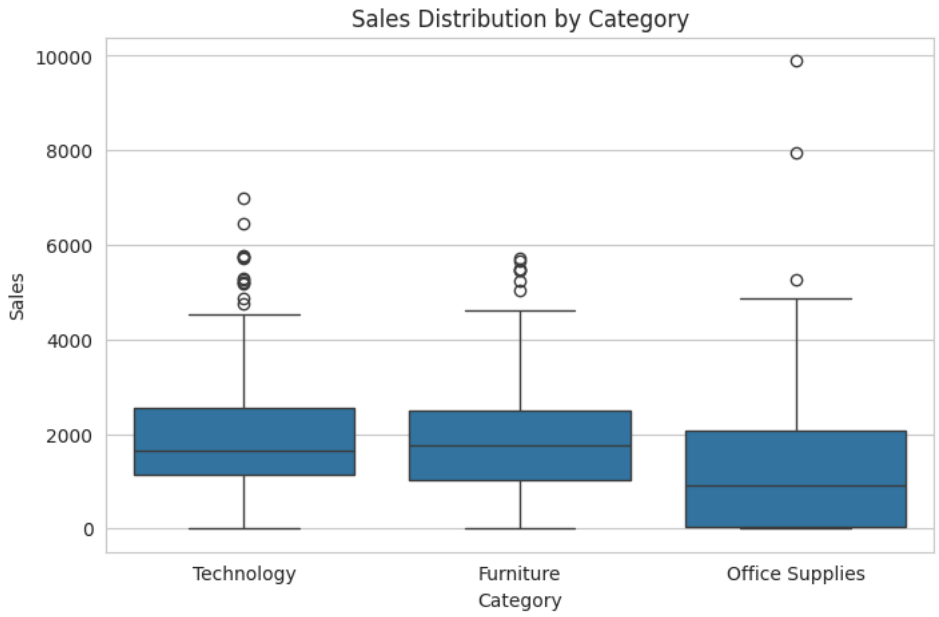
Step 6: Box Plot (Sales by Category)

```
plt.figure(figsize=(8,5))

sns.boxplot(data=df, x="Category", y="Sales")

plt.title("Sales Distribution by Category")

plt.show()
```



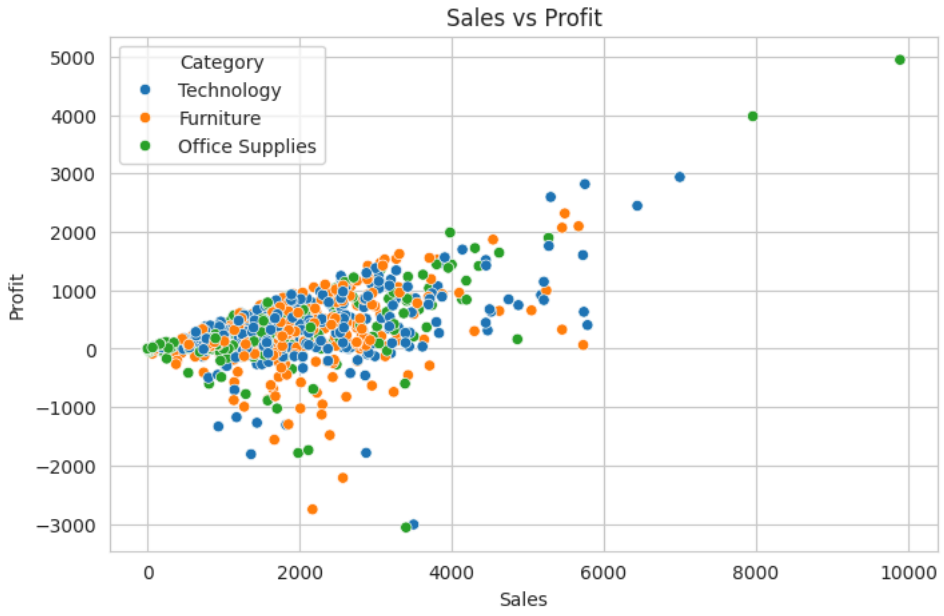
Step 7: Scatter Plot (Sales vs Profit)

```
plt.figure(figsize=(8,5))

sns.scatterplot(data=df, x="Sales", y="Profit", hue="Category")

plt.title("Sales vs Profit")

plt.show()
```



Step 8: Correlation Heatmap

Select numerical columns.

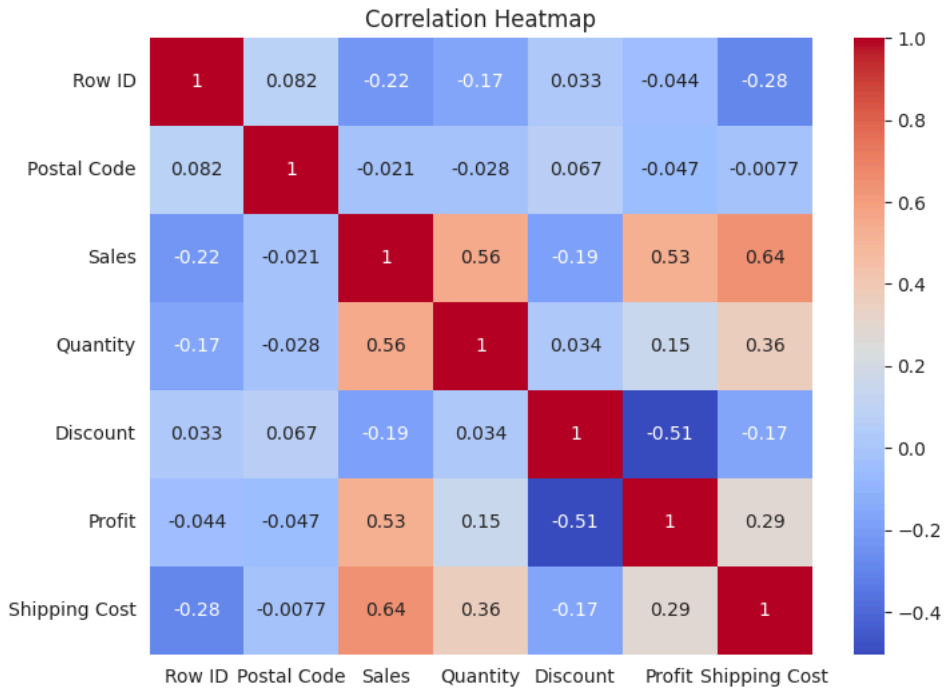
```
numeric_df = df.select_dtypes(include="number")

plt.figure(figsize=(8,6))

sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")

plt.title("Correlation Heatmap")

plt.show()
```



Step 9: Customized Visualization

Example with annotations.

```
sales_region = df.groupby("Region")["Sales"].sum()

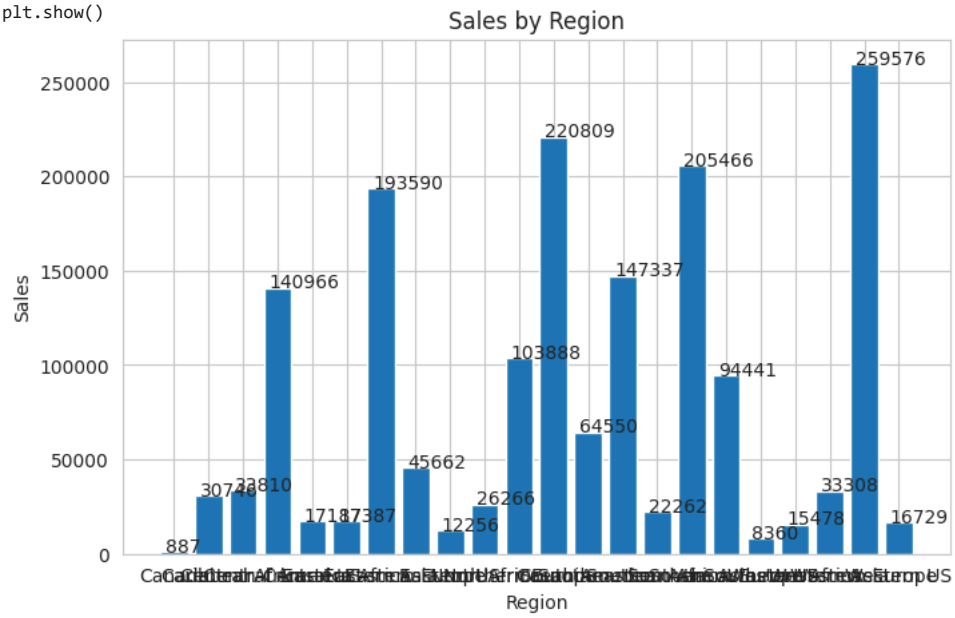
plt.figure(figsize=(8,5))

bars = plt.bar(sales_region.index, sales_region.values)

plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")

for bar in bars:
    plt.text(bar.get_x()+0.15,
             bar.get_height(),
             int(bar.get_height()))
```

```
plt.show()
```



Step 10: Business Insights

From the visualizations, we can conclude: Some regions generate significantly higher sales. SAles fluctuate over different months. Sales values are unevenly distributed. Certain product categories contain extreme outliers. Higher sales generally lead to higher profit, although exceptions exist. Correlation analysis helps identify related numerical features.

Conclusion

Advanced data visualization helps convert numerical data into meaningful insights. Charts such as bar plots, line charts, histograms, box plots, scatter plots, and heatmaps allow analysts to detect trends, identify outliers, understand relationships, and communicate findings effectively for better business decisions.

```
Start coding or generate with AI.
```